Lesson 7.1 - Continuous Probability Distributions

> **Video:** Continuous Probability Distributions - Basic Introduction by The Organic Chemistry Tutor **Chapter:** 7 - Continuous Distributions

Continuous random variables

A **continuous** random variable X can take **any** value in a range, such as 2, 3.5 or 4.68. A **discrete** one is restricted to specific values, such as 1, 2, 3, ...

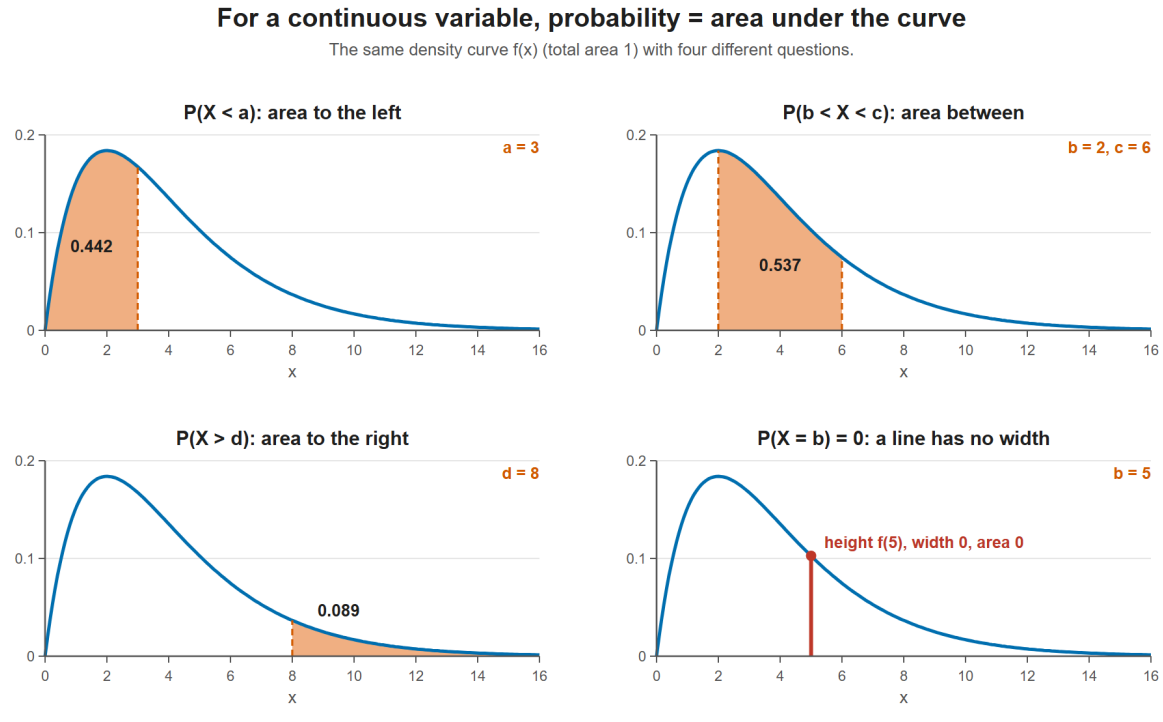
The curve of a continuous distribution is its **probability density function (PDF)**, $f(x)$. Its height above the x-axis at x is $f(x)$.

The key rules

1. $f(x) \geq 0$ everywhere: the curve never dips below the x-axis.
2. **Total area under the curve = 1.**
3. **Probability = area under the curve** over the interval you want:
   - $P(X < a)$ = area to the **left** of a
   - $P(b < X < c)$ = area **between** b and c
   - $P(X > d)$ = area to the **right** of d
4. **A single value has zero probability:** $P(X = b) = 0$. A vertical line has height but no width, so its area is 0.
5. Therefore $<$ and $\leq$ give the **same** probability:

$$P(X < a) = P(X \leq a), \qquad P(b < X < c) = P(b \leq X \leq c)$$

The **normal distribution** (bell curve) is the most important example. It is covered in Chapter 8.



Every probability question about a continuous variable is a shaded area under the density curve. A single value is a vertical line with no width, so its probability is 0.

Uniform distribution

$f(x)$ is **constant** between a and b, so the graph is a rectangle with area 1.

Example: a uniform distribution from $a = 2$ to $b = 6$ has base $= 4$, so the height must be $\frac{1}{4}$ for the area to equal $4 \times \frac{1}{4} = 1$.

PDF	Mean	Standard deviation
$f(x) = \frac{1}{b-a}$ for $a \leq x \leq b$	$\mu = \frac{a+b}{2}$	$\sigma = \frac{b-a}{\sqrt{12}}$

Exponential distribution

The curve starts at height λ on the y-axis and **decays** as x increases.

PDF	Rate	Left area	Right area
$f(x) = \lambda e^{-\lambda x}$	$\lambda = \frac{1}{\mu}$	$P(X < x) = 1 - e^{-\lambda x}$	$P(X > x) = e^{-\lambda x}$

Key takeaways

- Continuous distributions are described by a PDF $f(x) \geq 0$ with a total area of 1.
- **Probability = area under the curve.**
- $P(X = \text{one exact value}) = 0$, so "$<$" and "$\leq$" are interchangeable.
- Uniform: a flat rectangle at height $1/(b-a)$. Exponential: a decaying curve with $\lambda = 1/\mu$.

Lesson 7.2 - Cumulative Distribution Functions and Probability Density Functions

> **Video:** Cumulative Distribution Functions and Probability Density Functions by The Organic Chemistry Tutor
> **Chapter:** 7 - Continuous Distributions

PDF vs CDF

	PDF, probability density function	**CDF**, cumulative distribution function
Notation	$f(x)$	$F(x) = P(X \leq x)$
Tells you	The **shape** of the distribution and the curve's **height** at x	The **area to the left** of x, the probability accumulated up to x
Range	$f(x) \geq 0$	$0 \leq F(x) \leq 1$

The total area under any PDF is 1.

Uniform distribution on $[a, b]$

PDF (a constant height; no x in it):

$$f(x) = \frac{1}{b-a}$$

CDF (the area of the rectangle from a to x = base × height):

$$F(x) = P(X \leq x) = (x-a) \cdot \frac{1}{b-a} = \frac{x-a}{b-a}$$

Exponential distribution

PDF (it starts at height λ and decays, with $\lambda = 1/\mu$):

$$f(x) = \lambda e^{-\lambda x}$$

CDF (area to the left):

$$F(x) = P(X \leq x) = 1 - e^{-\lambda x}$$

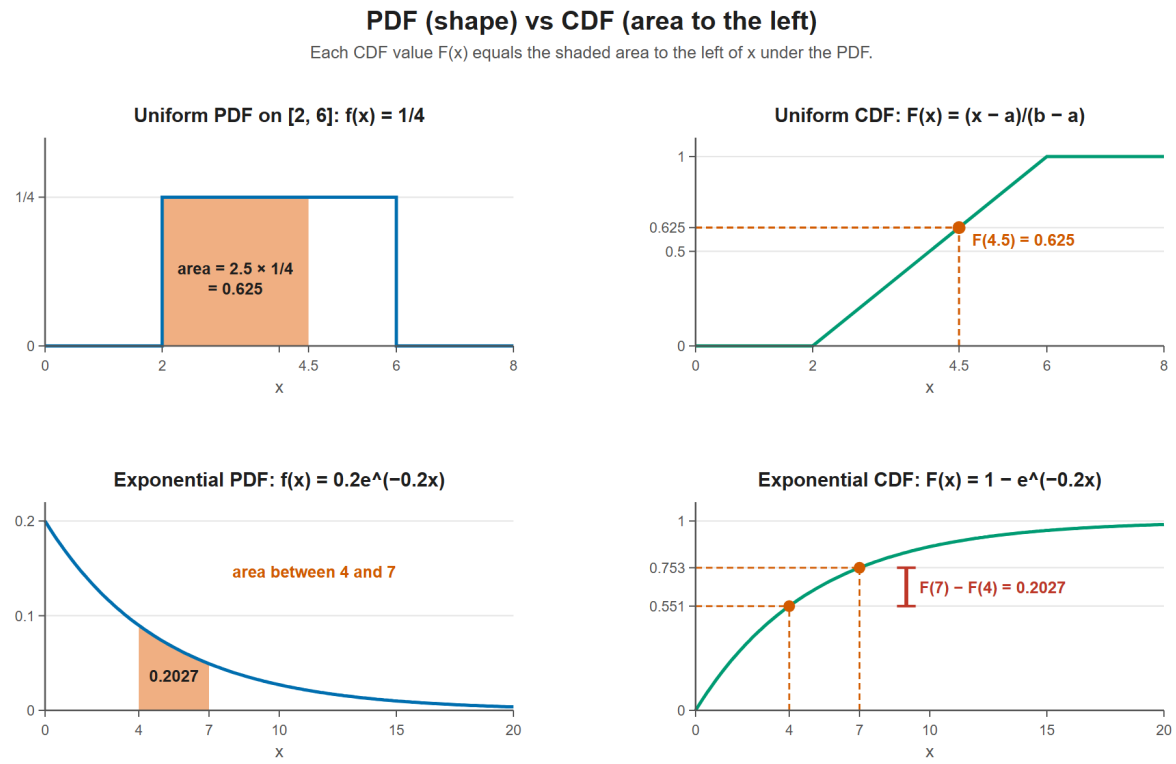
Area to the right (not a CDF, because the CDF is always the left area):

$$P(X \geq x) = 1 - F(x) = 1 - (1 - e^{-\lambda x}) = e^{-\lambda x}$$

Area between a and b: subtract two left areas.

$$P(a \leq X \leq b) = F(b) - F(a) = (1 - e^{-\lambda b}) - (1 - e^{-\lambda a}) = e^{-\lambda a} - e^{-\lambda b}$$

Picture: the region left of b minus the region left of a leaves the region between a and b.



Left column: the PDF, whose shaded area is a probability. Right column: the CDF, whose height at x is the area to the left of x. The area between 4 and 7 under the exponential PDF (0.2027) equals the jump F(7) − F(4) on its CDF.

Reminders for continuous distributions

- $P(a \leq X \leq b) = P(a < X < b)$, because the endpoints don't change the area.
- $P(X = a) = 0$: a single line has height but no width.

Key takeaways

- **PDF** $f(x)$ gives the **shape and height**. **CDF** $F(x)$ gives the **area to the left** (cumulative probability).
- Right area = $1 - F(x)$. Area between = $F(b) - F(a)$.
- Uniform: $F(x) = \dfrac{x - a}{b - a}$. Exponential: $F(x) = 1 - e^{-\lambda x}$.

Lesson 7.3 - Probability of a Uniform Distribution

> **Video:** Continuous Probability Uniform Distribution Problems by The Organic Chemistry Tutor **Chapter:** 7 - Continuous Distributions

Formulas (uniform on $[a, b]$)

The graph is a **rectangle** with total area 1:

$$\text{area} = \text{base} \times \text{height} = (b-a)\,f(x) = 1 \;\Rightarrow\; f(x) = \frac{1}{b-a}, \quad a \le x \le b$$

Quantity	Formula
Probability of an interval	**base × height** $= \dfrac{\text{interval width}}{b-a}$
Mean = median	$\dfrac{a+b}{2}$
Variance	$\dfrac{(b-a)^2}{12}$
Standard deviation	$\dfrac{b-a}{\sqrt{12}}$
k-th percentile value K	$\dfrac{k}{100} = \dfrac{K-a}{b-a} \;\Rightarrow\; K = a + \dfrac{k}{100}(b-a)$

Special cases:

- $P(X = c) = 0$ for any single value (a line has no width).
- $P(X > b) = P(X < a) = 0$ (the height is zero outside $[a, b]$).

Example 1 - waiting for a train (0 to 40 minutes)

The wait time is uniformly distributed on $[0, 40]$.

(a) PDF

$$f(x) = \frac{1}{b-a} = \frac{1}{40-0} = \frac{1}{40}, \qquad 0 \le x \le 40$$

(b) Graph: a flat line at height $\frac{1}{40}$ from $x = 0$ to $x = 40$, which forms a rectangle.

(c) Wait less than 8 minutes

$$P(X < 8) = \text{width} \times \text{height} = (8-0) \times \frac{1}{40} = \frac{8}{40} = \frac{1}{5} = 0.20 = 20\%$$

(d) Wait more than 30 minutes

$$P(X > 30) = \text{width} \times \text{height} = (40-30) \times \frac{1}{40} = \frac{10}{40} = 0.25 = 25\%$$

(e) Assorted probabilities

- Wait between 10 and 26 minutes:

$$P(10 < X < 26) = \frac{26-10}{40} = \frac{16}{40} = \frac{2}{5} = 0.40 = 40\%$$

- $P(X = 20) = 0$, a single value.

- $P(X > 45) = 0$, outside the range.

(f) Mean and standard deviation

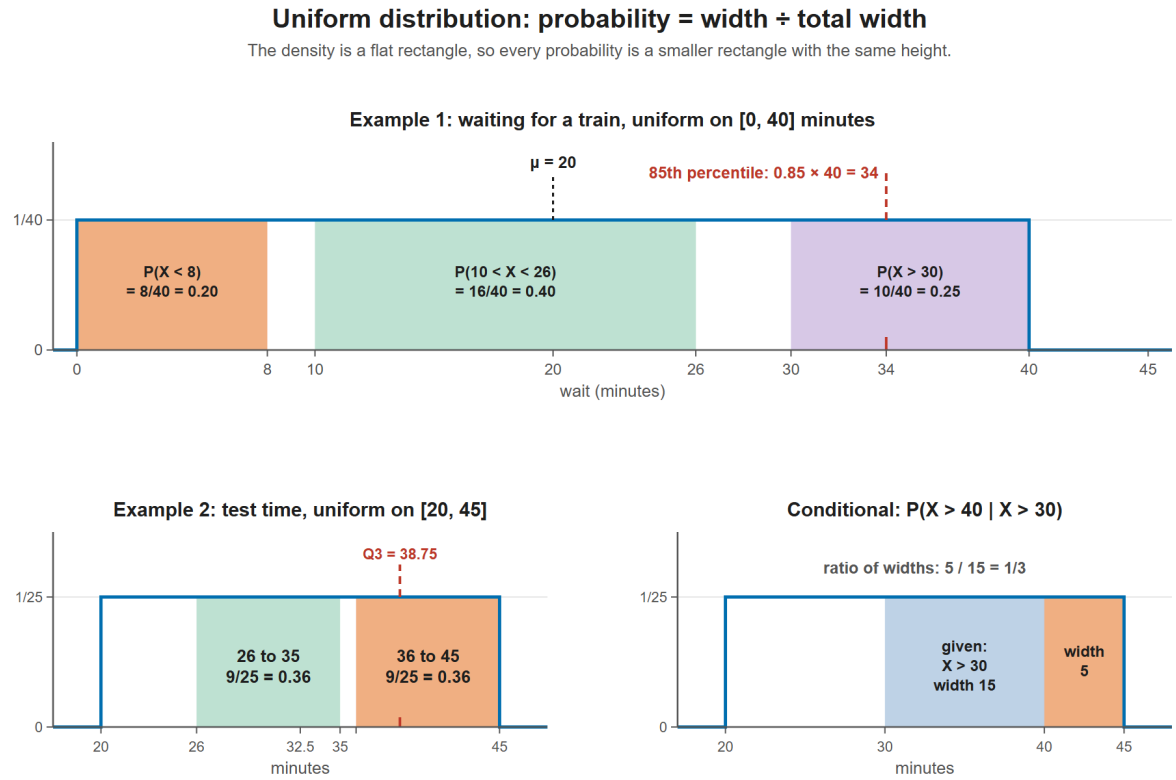
$$\mu = \frac{a+b}{2} = \frac{0+40}{2} = 20, \qquad \sigma = \frac{b-a}{\sqrt{12}} = \frac{40}{\sqrt{12}} \approx 11.55$$

(g) 85th percentile

The area to the left of K must be 0.85:

$$0.85 = (K-0)\cdot\frac{1}{40} \;\Rightarrow\; K = 0.85 \times 40 = 34 \text{ minutes}$$

This is reasonable: the 75th percentile is 30, so the 85th must lie between 30 and 40.



Each probability is the width of the shaded strip divided by the total width. Strips of equal width (26 to 35 and 36 to 45) have equal probability. For the conditional question, the given part (X > 30) becomes the new total width.

Example 2 - time to finish a chemistry test (20 to 45 minutes)

(a) PDF

$$f(x) = \frac{1}{b-a} = \frac{1}{45-20} = \frac{1}{25}, \qquad 20 \le x \le 45$$

(b) Graph: a rectangle of height $\frac{1}{25}$ from 20 to 45.

(c) More than 36 minutes

$$P(X > 36) = \frac{b-36}{b-a} = \frac{45-36}{25} = \frac{9}{25} = 0.36 = 36\%$$

(d) Between 26 and 35 minutes

$$P(26 < X < 35) = \frac{\text{width}}{b-a} = \frac{35-26}{25} = \frac{9}{25} = 0.36$$

The width is the same as in (c), so the probability is the same.

(e) Mean (= median), variance, standard deviation

$$\mu = \frac{a+b}{2} = \frac{20+45}{2} = 32.5, \qquad \sigma^2 = \frac{(b-a)^2}{12} = \frac{25^2}{12} = \frac{625}{12} \approx 52.08, \qquad \sigma = \sqrt{\sigma^2} \approx 7.217$$

(f) Third quartile (Q_3 = 75th percentile)

$$0.75 = (K-20) \cdot \frac{1}{25} \;\Rightarrow\; K - 20 = 18.75 \;\Rightarrow\; K = 38.75 \text{ minutes}$$

Check: halfway between the median 32.5 and the maximum 45 is $(32.5+45)/2 = 38.75$ ✓

(g) Conditional: more than 40 minutes, given more than 30 minutes

$$P(X > 40 \mid X > 30) = \frac{P(X > 40 \text{ and } X > 30)}{P(X > 30)} = \frac{P(X > 40)}{P(X > 30)}$$

If $X > 40$, then $X > 30$ is automatically true, so the intersection is just $X > 40$.

$$= \frac{(45-40) \cdot \frac{1}{25}}{(45-30) \cdot \frac{1}{25}} = \frac{5}{15} = \frac{1}{3} \approx 33.3\%$$

The heights cancel, so a conditional uniform probability is just a **ratio of widths**.

Key takeaways

- Uniform probability = **width of the interval ÷ total width**.
- A single point, or anything outside $[a, b]$, has probability 0.
- Percentile: solve $\frac{k}{100} = \frac{K-a}{b-a}$ for K.
- Mean = median = $(a+b)/2$, and $\sigma = (b-a)/\sqrt{12}$.

Lesson 7.4 - Probability of an Exponential Distribution

> **Video:** Probability Exponential Distribution Problems by The Organic Chemistry Tutor **Chapter:** 7 - Continuous Distributions

Formulas

Quantity	Formula
Rate parameter	$\lambda = \dfrac{1}{\mu}$
PDF	$f(x) = \lambda e^{-\lambda x},\ x \geq 0$
Area to the **left**	$P(X < x) = 1 - e^{-\lambda x}$
Area to the **right**	$P(X > x) = e^{-\lambda x}$
Area **between** a and b	$P(a < X < b) = P(X < b) - P(X < a) = e^{-\lambda a} - e^{-\lambda b}$

($e \approx 2.71828$. The video once says "rate parameter = 1/λ"; it means $\lambda = 1/\mu$.)

Example - laptop lifespan

Laptops from company XYZ last **5 years** on average, and the lifespan follows an exponential distribution.

(a) Rate parameter

$$\lambda = \frac{1}{\mu} = \frac{1}{5} = 0.20 \text{ per year}$$

(b) PDF and graph

$$f(x) = \lambda e^{-\lambda x} = 0.20\, e^{-0.20x}$$

The graph starts at the **y-intercept** $f(0) = 0.20 = \lambda$ (since $e^0 = 1$) and decays toward 0 as x grows.

(c) Lasts less than 3 years (left area)

$$P(X < 3) = 1 - e^{-\lambda x} = 1 - e^{-0.20(3)} = 1 - e^{-0.6} \approx 0.4512 = 45.12\%$$

(d) Lasts more than 10 years (right area)

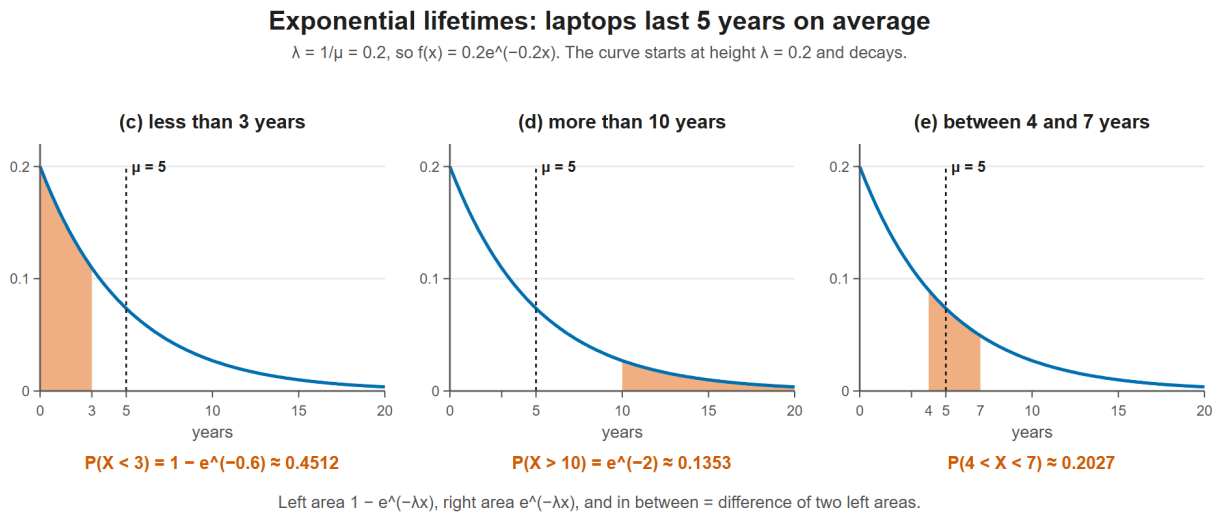
$$P(X > 10) = e^{-\lambda x} = e^{-0.20(10)} = e^{-2} \approx 0.1353 = 13.53\%$$

(e) Lasts between 4 and 7 years

The area left of 7 minus the area left of 4 leaves the region between them:

$$P(4 < X < 7) = P(X < 7) - P(X < 4) = \left(1 - e^{-1.4}\right) - \left(1 - e^{-0.8}\right) \approx 0.75340 - 0.55067 = 0.20273 \approx 20.3\%$$

(Equivalently, $e^{-0.8} - e^{-1.4} \approx 0.4493 - 0.2466 = 0.2027$.)



The three laptop questions as areas under f(x) = 0.2e^(−0.2x). The curve starts at λ = 0.2, and the mean lifetime is 5 years.

Key takeaways

- Find $\lambda = 1/\mu$ first. The PDF's y-intercept equals λ.
- Left area $1 - e^{-\lambda x}$; right area $e^{-\lambda x}$; between = difference of the left areas.
- The exponential distribution models **waiting times and lifetimes**. Short times are most likely, and long times become less and less likely.